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High quality remote laser cutting of polymer lithium-ion battery separator using a NIR picosecond-pulsed laser

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ABSTRACT

Due to the increasing demand for electrification in general and battery manufacturing in particular, there is a need to use digital tools wisely. Laser is a digital tool that could process materials with high precision and quality. The polymer separator is one of the basic components of lithium-ion battery, which has the significant effect on the performance of lithium-ion battery. The remote laser cutting of the 25 μm-thick lithium-ion battery separator was experimentally investigated using nanosecond and picosecond near-infrared laser systems. The ns-pulsed laser was found to be sensitive to the material directionality and it could not achieve melt free cuts at processing conditions suitable for the industrial production. The ps-pulsed laser produced clean cut kerfs at high productivity. In terms of cutting productivity, the cutting speed of the ultrashort pulse laser system is approximately 700 times that of the short pulse laser system. To estimate the change in the material's optical properties, the fluence efficiency model was employed. It was found that the apparent optical absorptivity of the material changes from 0.4 cm^{-1} to 1400 cm^{-1} when the pulse duration is changed from the nanosecond to the picosecond regime for the same wavelength.

1. Introduction

The demand for electric vehicles in the upcoming years pose significant challenges in terms of the new components to be manufactured [1]. The increasing energy storage demand calls for rapid and robust manufacturing techniques for the electric mobility sector. Cathode, anode, and polymer separator are the basic elements of a lithium-ion battery (LIB). The manufacturing process of LIBs involves many different steps, with the productivity and quality of each step critical to the overall quality of the battery. The current state-of-the-art battery manufacturing process comprises three main stages: electrode preparation, cell assembly and activation of the battery's electrochemistry. First, the active material, conductive additive and binder are mixed with a solvent to form a uniform slurry. This slurry is transferred to a slot die and coated on both sides of the current collector. Aluminium foil is used for the cathode, and copper foil is used for the anode. The coated sheet is then conveyed to drying equipment to remove the solvent. Subsequent calendaring can be used to adjust the physical properties (such as adhesion, conductivity, density, and porosity) of the electrodes. Following these processes, the finished electrodes are notched and cut to the required dimensions to fit the cell design. The electrodes are then placed in a vacuum oven to remove excess moisture. The moisture content of the electrodes is checked after drying to minimise side reactions and corrosion within the cell. Once the electrodes are adequately prepared, they are transported to a dry room with separators for cell assembly. The LIB anode, cathode, and separator are each manufactured on separate production lines. The electrodes and separator are

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stacked in layers to form the internal structure of a cell. Aluminium and copper tabs are welded to the cathode and anode current collectors, respectively. The cell stack is then transferred to a special casing. The casing is filled with electrolyte before final sealing, completing the cell manufacturing process. Prior to delivery to product manufacturers, the cells undergo electrochemical activation steps to ensure operational stability [2].

The separator of LIBs is commonly made of microporous propylene [3,4]. The dry process is one of the dominant methods for the large-scale production of microporous separators for LIBs. In this process, polyolefins are melted, extruded into a sheet, and annealed to induce crystallite formation. This is followed by uniaxial stretching to form the microporous structure. The microporous structure is formed by tearing apart stacked lamellar crystallites and is oriented in the machine direction (MD). The uniaxial stretching leads to a high anisotropy in the mechanical properties, resulting in membranes with much higher tensile strength in the MD than in the transverse direction (TD). However, the thermal shrinkage in the TD is negligible. The porous structure of the separator is strongly influenced by the morphology of the extruded film as well as the processing parameters such as stretching ratio, speed, annealing temperature, and time [5]. Puncture resistance is used to indicate the tendency of the separator to allow the short circuit to occur during battery assembly. In other words, the puncture resistance indicates the resistance of the separator to local force. In addition, the puncture resistance indicates how resistant the separator is to breakage by dendrite formation. On the other hand, the separator must not shrink significantly in the operating temperature range of LIBs. Therefore, separator shrinkage at a given temperature is an important factor in evaluating the separator. Porosity affects the ability to tap sufficient electrolyte in the pores and maintain high ionic conductivity. However, the distribution and size of the pores should be noted as they affect the performance of the LIBs [5].

The most basic type of microporous membrane separator is the monolayer membrane, which can be made from a variety of polymeric materials. Polyolefins are commonly used for monolayer membrane separators due to their superior mechanical strength

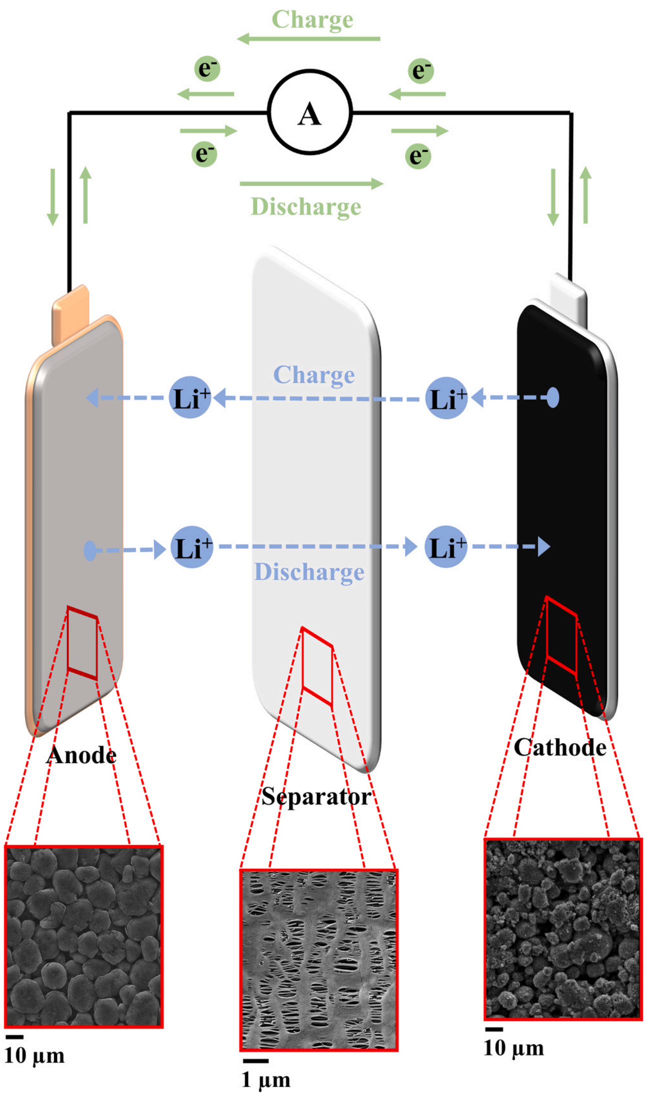

Fig. 1. Graphical representation of anode, cathode, and separator in the lithium-ion battery.

and chemical stability. The thickness of monolayer separators ranges from 13 to 300 µm [6]. In the state-of-the-art process, LIB components are punched using dies or trimmed by a cutting blade or by repeated punching [7].

Fig. 1 shows the schematic diagram of the working principle of the LIB and SEM images of the anode, cathode, and separator. When the LIB battery is discharged, the lithium-ion intercalated in the anode active material moves into the electrolyte. It then passes through the porous polymer separator and reaches the cathode active materials. The separator is made of a porous polymer material and plays a critical role in the LIB's performance. It enables lithium ions to pass between the cathode and anode through its electrolyte-filled pores. At the same time, it prevents short circuits between the electrodes. Although the separator is not an active component in the electrochemical performance of LIBs, it is critical to lithium-ion transport, rate performance, cell life and overall battery safety [6].

The manufacture of LIBs relies on mechanical punching methods to process the electrodes. In this process, the precision of the cut is critically dependent on the gap between the punching tools. To achieve high quality cuts, the tool gap must remain within 3–4 % of the thickness of the electrode sheet. For example, when cutting a 100 µm anode, the gap must be kept below 4 µm. However, such tight tolerances significantly increase tooling and maintenance costs. In addition, frequent changes in electrode chemistry and dimensions during production require adaptable manufacturing techniques [8]. On the other hand, the rapid growth of renewable energy systems and electric vehicles over the past decade has led to an unprecedented demand for LIBs. While LIBs are widely preferred for electronic devices due to their high energy density and reliability, their scalability in mass production further strengthens their market dominance. Despite predictions that LIBs will remain the leading energy storage technology for the next decade, the expansion of large-scale battery manufacturing facilities (Giga-factories) faces persistent hurdles. Key challenges include the complexity of the manufacturing processes and the large amount of machinery required, which are hindering industry-wide adoption of end-to-end battery manufacturing solutions [9]. In one hand, remote laser cutting has emerged as a promising alternative, offering greater flexibility for dynamic production requirements of battery production [8]. On the other hand, it could satisfy requirement of mass production through the higher productivity and quality. As the trend in manufacturing is towards digital manufacturing, the laser could be a promising tool due to its non-contact nature and ease of integration. Lasers for industrial manufacturing purposes could be generically classified into different categories based on the temporal distribution of the emitted laser and the wavelength of the emission. Concerning laser system, the ns-pulsed laser constitutes arguably the most economical choice in terms of microcutting operations. The short-pulsed regime corresponding to nanosecond range provides relatively moderate cutting speeds on metals with moderate cut quality. Picosecond lasers are in the longer range of ultrashort-pulsed regime if the pulse duration is in the order of typically 10 ps, where the quality can be improved significantly. Femtosecond pulses produce better quality on metals but require a higher capital cost. On the other hand, the effect of the pulse durations on separator materials requires experimental work.

The absorptivity of dielectric materials differs from that of metals depending on the wavelength. The CO_2 laser sources operating at infrared (IR) wavelengths are commonly used for polymer processing [10]. Caiazzo et al. [11] investigated CO_2 laser cutting of polyethylene, polypropylene and polycarbonate of various thicknesses ranging from 2 to 10 mm. Davim et al. [12] investigated laser cutting of polymethylmethacrylate, polycarbonate, polyethylene, and reinforced thermoset plastics. According to their findings, the laser cuttability is ranked as follows: polymethylmethacrylate - very high, polycarbonate - high, polyethylene - high to medium and reinforced thermoset - lower. Khoshaim et al. [13] investigated the effects of four CO_2 laser parameters on quality. The works show that IR lasers generate a thermal separation process where cut kerfs are characterized by molten material often accompanied by the presence of burr. Improved process quality during the laser micromachining of polymer films can be achieved using a third-harmonic Nd:YAG laser in UV. Polymeric materials have high absorptivity at the UV wavelength as well as inducing photochemical material decomposition [14] that enables precise singulation, micro-drilling and the creation of deep micro-grooves. Thermoplastic films of various thicknesses are used to produce these microstructures, the quality of which is evaluated by optical and scanning electron microscopy [15,16]. UV lasers offer high precision. However, their lifetime is relatively short due to the wear of the harmonic crystal and optical components in the laser source. Moreover, significant power losses are present with harmonic generations. The near infrared (NIR) laser sources based on fiber and diode lasers are appealing solutions for industrial manufacturing processes due to their high stability and reduced capital cost. However, concerning polymeric material processing the suitability of NIR laser sources depends mainly on the optical properties of the material [17] as often these materials are highly transparent in this wavelength range [18,19].

In addition to the wavelength, the temporal distribution of the emitted laser has an important role in the processing of dielectric materials. The use of ultrafast lasers can enable the processability of polymeric materials at NIR wavelengths despite the transparency of the material in room temperature conditions [20,21]. Rahaman et al. [9] developed an analytical solution for a transient two-dimensional thermal model to investigate the influence of laser parameters during ultrafast laser interactions with polypropylene, a material widely used in industrial applications. To validate the theoretical calculations, laser cutting experiments were performed on polypropylene sheets. The study identified total energy absorption and laser intensity as key factors in evaluating laser processing performance. Almeida et al. [22] studied the generation of 2000, 6000 and 10000 pores per cm^2 on the surface of natural rubber latex using a femtosecond (fs)-pulsed laser system. Another application of ultrashort laser systems is the cutting of photo deformable liquid crystal polymers, which undergo geometric deformation when exposed to light. These materials are critical for smart and multi-functional applications due to their compact size, fast response, remote controllability, high integration, and versatile functionality. Wang and Qian [23] developed a fs laser direct writing method for the high-resolution, programmable fabrication of arbitrarily patterned photochemical and photothermal effect-based photo deformable liquid crystal polymer micro actuators.

Ultrashort pulse laser systems could produce peak intensities in the range of several TW/cm^2 due. This high intensity facilitates non-linear absorption of pulse energy, allowing precise laser-induced breakdown thresholds at reduced pulse energy [24,25]. Such properties make these systems well suited for applications such as cutting transparent glass [26]. In addition, the non-linear absorption properties of ultrashort pulses enable micromachining of three-dimensional structures within the bulk of transparent materials. Itoh et al. [27] reviewed the mechanisms and techniques for bulk modification of transparent materials using femtosecond laser pulses, discussing the fabrication of photonic and other structures, such as waveguides, couplers, gratings, diffractive lenses, optical data storage systems, and microfluidic channels. While proving to be an effective tool for processing transparent materials, it has been shown that heat accumulation may occur and can be detrimental on the process quality when high pulse repetition rates and high energy levels are used with the ultrafast lasers [28,29]. The balance of pulse energy and the pulse repetition rate is essential for improving both the productivity and the quality in the ultrafast laser processing.

A key advantages of ultrashort pulse laser systems for electric mobility applications is their versatility in processing different materials with limited thermal damage. These systems have been investigated for cutting LIB cathodes [30], anodes [31], and lithium metal used as an anode in solid-state batteries [32]. The ultrashort pulse laser systems offer the possibility of using the burst mode [33]. The use of burst mode showed to be effective in increasing process productivity along with quality [34]. The same laser source can be potentially adapted to the cutting of the separator material at NIR wavelengths without the need for harmonic conversion. Since in the battery production, there is a need for the robust laser system in which it could wide range of materials [8]. The selection of laser source for this requirement seems to be essential to respond the flexibility in the material processing. The main objective of this research work is to identify the suitable laser source for cutting LIB separator and optimize the cutting process experimentally. To the authors' knowledge, there is no publication in the literature on laser cutting of LIB separators. Accordingly, this work compares the use of nanosecond (ns)- and ps-pulsed laser systems at near-infrared wavelengths for the laser cutting of LIB separators. The work shows experimentally the change of optical absorption behaviour as a function of the pulse duration as observed in the different processing regimes with the ns- and the ps-pulsed laser sources. The fluence efficiency model was employed using the cut conditions with the ps-pulsed laser systems to estimate the effective optical absorption. The Beer-Lambert model was then used to explain the difference in cutting speed between the ultrashort and short pulse systems.

2. Modelling

2.1. Modelling of light absorption and transmission

Reflectivity and absorptivity properties of a material are related to the complex refractive index:

$$\widetilde{n} = n + ik \tag{1}$$

where n is the real part of the refractive index and k is the imaginary part, called the extinction coefficient. The absorption coefficient is thus calculated from the extinction coefficient for a given wavelength (λ) through the following relationship:

$$\alpha = \frac{4\pi k}{\lambda} \tag{2}$$

Correspondingly, the optical penetration depth can be calculated as the inverse of the absorption coefficient.

$$\delta = \frac{1}{\alpha} \tag{3}$$

The transmitted energy on the workpiece (T) can be calculated omitting the reflections at the interfaces and only considering the attenuation according to the Beer-Lambert law from the following expression:

$$T = I_0 e^{-\alpha z} \tag{4}$$

where I_0 is the incident light intensity of light and z is the distance travelled in the medium. The optical properties of solids vary as a function of temperature ($Temp$) with $\widetilde{n} = f(Temp)$. When the beam intensity is above a certain threshold of changing material opacity, the apparent optical behaviour can be observed as an increase in the absorption coefficient and hence a reduction in the optical penetration depth. In a simple approximation, the high peak intensity of the ultrashort pulses creates the necessary thermal conditions leading to improved machining behaviour as $T = f(I_0)$. The transverse mode of both laser systems used in the experiment is TEM_{00}. This transverse mode corresponds to the Gaussian distribution. The square pulse shape assumption is considered to calculate the peak intensity of a single pulse. For a square shaped pulse, the peak intensity can be calculated as

$$I_0 = \frac{8P_{pk}}{d_0^2\pi} = \frac{8E}{\tau d_0^2\pi} \tag{5}$$

where P_{pk} is the peak power, d_0 is the beam diameter, E is the energy of pulse and τ the pulse duration. The pulse duration is generally a specific characteristic of the laser source, whether it is in the long (ms-μs), short (ns), or ultrashort (ps-fs) regime. Accordingly, the overall change in absorptivity can be achieved mainly by moving towards ultrashort pulses, which generate a higher intensity of the beam. Therefore, it can be said that the optical apparent optical penetration depth ($\widehat{\delta}$) and hence the apparent optical absorption ($\frac{1}{\widehat{\alpha}}$) can be estimated different as a function of the pulse duration as:

$$\widehat{\delta} = \frac{1}{\widehat{\alpha}} = f(\tau) \tag{6}$$

The apparent optical penetration depth can be observed empirically as an improvement in machining behaviour, such as the separation of the material at a wavelength that is transparent to the solid at room temperature. It can also be estimated by using the ablation efficiency model as described below.

2.2. Estimation of the optical absorptivity through ablation efficiency

The relationship between the material removed per a single pulse and the optical absorption can be established using the ablation efficiency model [35]. The model estimates the behaviour of the volume removed per a single pulse and the peak fluence showing a peak value at the optimal peak fluence. The energy of each pulse for the burst mode can be calculated using the following formula.

$$E = \frac{P_{avg}}{PRR.Nb} \tag{7}$$

where Nb is the number of pulses in the burst, P_{avg} is the average power of laser, and PRR is the pulse repetition rate. For the laser system without the burst mode Nb is equal to 1. The peak fluence for the Gaussian spatial distribution is calculated using the following formula

$$\phi_0 = \frac{8.E}{\pi.d_0^2} \tag{8}$$

where d_0 is the diameter of the laser beam at focal point, and ϕ_0 is the peak fluence. It should be noted that the peak fluence represents the irradiated energy of a single pulse in the centre of the Gaussian distribution. The dependence of the volume removed by a single pulse can be described by the following formula.

$$\frac{V}{E} = \frac{1}{2}.\frac{\widehat{\delta}}{\phi_0}.\ln^2(\frac{\phi_0}{\phi_{th}}) \tag{9}$$

Where V is the volume removed by a single pulse, $\widehat{\delta}$ is the apparent optical penetration depth of a single pulse, and ϕ_{th} is the damage threshold of the material. Based on the literature, increasing the peak fluence increases the specific volume removal up to a certain threshold, after which the volume removed by each pulse starts to decrease. This trend shows that there is an optimum value of peak fluence to maximise the volume of material removed by the ultrashort pulse laser system. This optimum value is e^2 time the damage threshold. The damage threshold typically depends on the laser and material characteristics.

The model has been used for laser micromachining applications where the pulses overlap in the scan direction and the successive scan trajectories produce a finite depth of material removal. In the case of laser micro cutting, there is a complete separation across the entire thickness t of the material being machined, which can result in a loss of laser energy through the open kerf. However, the model can be used to evaluate the efficiency of different machining conditions that produce fully separated kerfs. To calculate the volumetric material removal rate for the remote laser cut sample, the sample thickness should be multiplied by the kerf width w_k and the cutting speed v_c of the sample. By dividing the volumetric material removal rate by the PRR and Nb, we can calculate the volume of the material removed by a single pulse.

$$\frac{V}{E} = \frac{t.w_k.v_c}{PRR.Nb.E} \tag{10}$$

The mathematical formulation shown in Eq. 9 can be fitted using the calculated volume removed and the peak fluence values using non-linear regression to estimate the damage threshold and the apparent optical penetration depth. The fitted apparent optical

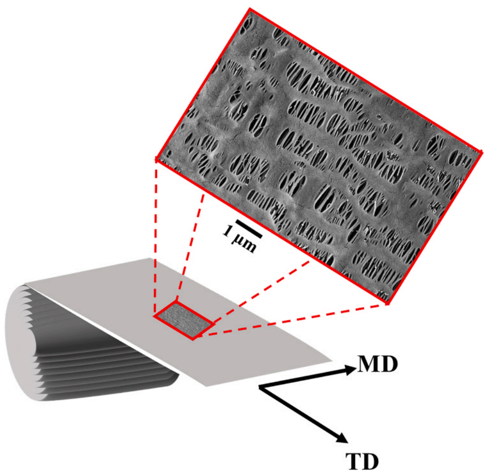

Fig. 2. Graphical representation and SEM image of LIB separator.

penetration depth value can be compared with the tabulated data of the material measured at the same wavelength under the room temperature conditions.

On the other hand, pulse overlap (*P.O.*) is another parameter that is a key parameter in the laser cutting operations. The distance between pulses on the sample depends on the scanning speed of the laser beam, the repetition rate, and the calculated spot diameter of the laser at the focal point. It can be calculated using the following formula.

$$P.O. = \left(1 - \frac{v_c}{d_0.PRR}\right).100 \tag{11}$$

3. Materials and methods

3.1. Separator material

The Celgard 2500 Monolayer Microporous Membrane separator, made of polypropylene with a thickness of 25 μm, was used in this study for cutting with two laser systems. Fig. 2 shows the microporous shape and orientation based on the MD and TD. The characteristics of this polymeric separator are given in Table 1. The pore of the separator is oval in shape and the dimension of these pores is different in MD and TD. The size of 30 holes in two directions was measured based on the SEM image of Fig. 2. The average value for two directions of hole size is given in the following.

The optical properties of the material make it very difficult for laser processing. The material is highly transparent and has very low absorptivity of 0.4 cm^{-1} at the 1064 nm wavelength of the conventional fiber lasers. The absorptivity at 10600 nm of the CO_2 lasers is relatively higher at 1.5 cm^{-1} [36]. On the other hand, the porous structure of the material gives it anisotropy in the TD and MD directions, which is expressed by the different shrinkage behaviour. Although the pore sizes are smaller than the laser beam sizes used in industrial operations (20–40 μm), the optical and thermal behaviour can be expected to vary as a function of the direction during the cutting process.

3.2. Laser cutting systems

Two different laser systems were used to cut the LIB separator. The first system uses a ns-pulsed laser, while the second system uses a ps-pulsed laser with burst mode capability. Both systems operate in the near infrared wavelength (1064 nm), and their technical specifications are detailed in Table 2. The ns-pulsed laser system was a 250 ns Q-switched fibre laser (YLP-1/100/50/50 from IPG Photonics, Oxford, MA, USA). The laser was coupled to a scanner head (TSH 8310 from Sunny Technology, Beijing, China) equipped with a 100 mm focal length f-theta lens (SL-1064–70–100 from Wavelength Opto-Electronic, Ronar-Smith, Singapore). With this system, scanning speeds of up to 3 m/s could be achieved. The laser produced an average power of up to 54 W and operated within a pulse repetition rate range of 20–80 kHz.

The second laser system, with a pulse duration of 10 ps and a maximum average power of 50 W (Picoblade 2, Lumentum, Schlieren, Switzerland), supports a wide range of pulse repetition rates and has burst mode capability with an intra-burst delay of 12.2 ns. The collimated beam of the laser, expanded to 15 mm, was directed into a galvanometric scanning head (Miniscan III-20, Raylase, Weßling, Germany) and focused by an f-theta lens (4401–288–000–20, Qioptiq, Göttingen, Germany) with a focal length of 254 mm. The calculated beam diameter at the focal point was 27.5 μm. The peak power of each pulse within the burst is equal and was characterised using a fast photodiode (DET10A2, Thorlabs, Newton, New Jersey, United States), which converts light energy into electrical current with a rise time of 1 ns. A digital oscilloscope (MSO 4024, Rigol, Beijing, China) was used to record the signals. In addition, the kerf widths of the cuts were examined using optical microscope (OM) images taken with the (echoLAB UM 300I BD, Paderno Dugnano, Italy).

3.3. Experimental design

An experimental procedure was carried out using two laser systems operating in the ns and ps pulse duration range to evaluate the processability window for laser cutting of LIB separators. For both laser systems, a dedicated experimental campaign was designed to

Table 1
Physical property provided by the Celgard and optical property of separator [36].

Parameter	Value
Thickness, t (μm)	25
Porosity	55 %
TD shrinkage at 90 °C	0 %
MD shrinkage at 90 °C	1.1 %–5.05
TD tensile strength (kg/cm^2)	80–130
MD tensile strength (kg/cm^2)	770–1170
Puncture strength (gf/cm^2)	200–325
Optical absorptivity at 1064 nm, $\alpha_{1064\ nm}$ (cm^{-1})	0.4
Optical absorptivity at 10600 nm, $\alpha_{10600\ nm}$ (cm^{-1})	1.5

Table 2
General specifications of ps- and ns-pulsed laser systems.

Parameter	ns-pulsed	ps-pulsed
Wavelength, λ (nm)	1064	1064
Max. average power, P_{avg} (W)	54	50
Pulse repetition rate, PRR (kHz)	20–80	200–8200
Beam quality factor, M^2	1.7	1.2
Polarisation state	Random	Linear
TEM mode	TEM_{00}	TEM_{00}
Collimated beam diameter, *dc* (mm)	5.9	15
Focal length, *f* (mm)	100	254
Calculated diameter at focal point, d_0 (μm)	39	27.5

match its characteristics. Each laser system was analysed individually, taking into account its specific operating parameters. Fig. 3 shows the experimental plans carried out for both laser systems, focusing on average power and peak fluence. The focal point of both laser systems was fixed at the top of the workpiece. Table 3 reports the fixed and variable process parameters used in this study.

For the experimental campaign with the ns-pulsed laser system, three levels of average power (24, 36 and 53 W) and two levels of pulse repetition rate (50.0 and 80.0 kHz) were selected to cover a wide range of peak fluence and pulse overlap. Laser cutting of the separator was performed in two different directions – MD and TD – to investigate the effect of pore orientation on laser cutting. The lines were scanned with the scanner head, increasing the speed in 0.005 m/s increments to complete the feasibility window. The line that achieved complete separation at the highest speed was selected for further investigation.

For the ps-pulsed laser experimental design, the average laser power was fixed at 31 W. Eight levels of burst number (*Nb*, ranging from 1 to 8) and three levels of pulse repetition rate (200.0, 550.3 and 901.0 kHz) were varied. The burst profile consisted of pulses with equal energy and peak power. According to Eq. 7, the energy of a single pulse decreases as the number of Nb and PRR increases. For example, the highest peak fluence in Fig. 3 belongs to Nb = 1 and PRR = 200.0 kHz. On the contrary, the lowest peak fluence of a single pulse in the ps-pulsed laser system belongs to Nb = 8 and PRR = 901.0 kHz. These parameter sets have been chosen for the ps-pulsed laser system to cover a wide range of peak fluence of a single pulse within a burst. For each set of process parameters, different lines were scanned by the scanner head at 0.5 m/s increments. The line that achieved complete separation of the workpiece was selected for further investigation. Samples with the highest cutting speed were then used to fit the ablation efficiency model. Laser cutting of the separator was performed in the MD and TD using the ps-pulsed laser system. This work aims to show the use of ps-long pulses changes the laser-matter interaction for the cutting process rather than providing the influence of pulse duration with the same energy content. This requires benchmarking the distinct laser technologies with their available characteristics.

4. Results

4.1. Cutting with ns-pulsed laser system

Fig. 4 shows the processability window for both directions. No separation of the sample is observed in the MD. The results indicate a relationship between the direction of the pores in the separator and laser cutting with the ns-pulsed laser system. This poses a challenge for scaling up laser cutting of separators in an industrial environment. The issue arises because the maximum cutting speed varies

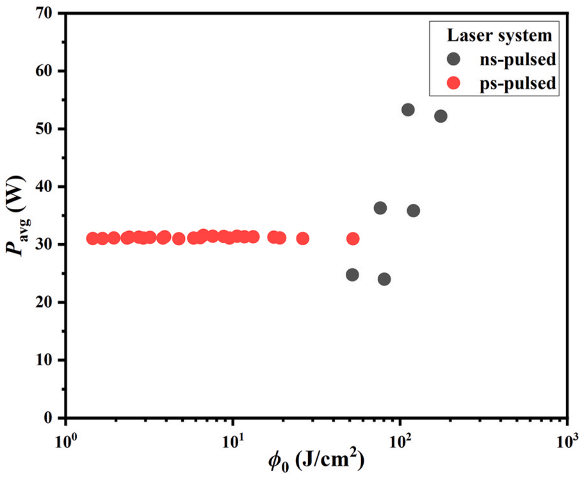

Fig. 3. Average power of the ns- and ps-pulsed laser systems versus the peak fluence for the laser cutting of separator explored in the experimental campaign.

Table 3
Process parameters variation of experimental campaign.

Parameter	ns-pulsed	ps-pulsed
Average power, P_{avg} (W)	24, 36, 53	31
Focal point, Δz (mm)	0	0
Number of pulses within a burst, Nb	NA	1–8
Pulse Repetition Rate, PRR (kHz)	50.0, 80.0	200.0, 550.3, 901.0
Cutting direction	MD, TD	MD, TD

depending on the cutting direction. In addition, the cutting speed is much slower than the industrial target of 1 m/s.

Fig. 5 shows the optical microscopy image of the samples at the average power of 36 W and two different scanning speeds on the TD. As shown in Fig. 5, melting starts when the scanning speed is low. As the laser beam progresses during scanning, the melting region enlarges. This indicates significant lateral thermal diffusion. Significant colouration is observed around the cut kerf and particularly at the final tip of the cut kerf. This point is visible in the figure at higher magnifications of the sample at the lower scanning speed. On the other hand, when the scanning speed is increased, only superficial melting occurs. In this case, no separation of the sample is observed. The images suggest that the laser cutting of the separator with the ns-pulsed laser system is based on melting assisted by local burning of the material.

4.2. Cutting with the ps-pulsed laser system

During the laser cutting with the ps-pulsed laser system, no significant difference in material direction was observed. Therefore, the results are presented excluding the effect of the material direction. Fig. 6. depicts the process feasibility window. Eight levels of number of pulses within the burst have been selected (1, 2, 3, 4, 5, 6, 7, 8). Three levels of repetition rates were chosen for the experimental design (200.0, 501.3, 901.0 kHz). According to Eq. 7, By increasing the repetition rate and the number of pulses within the burst, the energy of the pulses (or the peak fluence of each pulse) is reduced. For example, the highest peak fluence corresponds to Nb = 1 and PRR = 200.0 kHz, and the lowest peak fluence corresponds to Nb = 8 and PRR = 901.0 kHz in this study. For example, for PRR = 200.0 kHz, by increasing the number of pulses within the burst, the maximum cutting speed (threshold cutting speed between the Cut and No Cut zones) is changed. This means that the maximum cutting speed of the samples is linked to the energy of a single pulse within the burst. The same trend can be seen for the other repetition rates.

Fig. 7 shows the evolution of the cut kerf formation with ps-pulsed laser pulses at speeds of 6.0, 7.5 and 8.5 m/s (with the same number of the pulses in the burst and 200 kHz repetition rate) but different scanning speed. At a scanning speed of 8.5 m/s, a series of circular holes are produced with an average diameter of 34.1 ± 1.3 μm (mean ± standard deviation). Optical microscopy images show that the centre of the holes has been completely ablated, and the circumference of the holes has been melted. This occurs because the circumference of the hole is affected by the tail of the Gaussian energy distribution. The distance between the bursts can be calculated from the repetition rate of the burst and the scanning speed. If the scan speed is 8.5 m/s, the distance between two bursts is 42.5 μm. As a result, the wall thickness between two holes is 8.4 μm. Reducing the scan speed creates the conditions are set for reducing the wall thickness between two successive holes.

At a scanning speed of 7.5 m/s, the thickness of this wall is approximately 3.4 μm. At this scanning speed, the holes are not perfectly circular. As shown in Fig. 7, the shape of the hole is a circle with a small bulge. The dimension of these bulges has been measured for 15

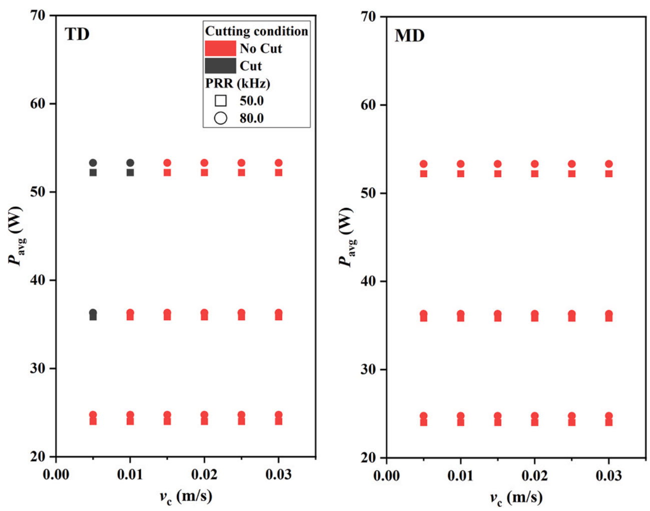

Fig. 4. Process feasibility window for the ns-pulsed laser cutting, shown as a function of average power, cutting speed, and material orientation.

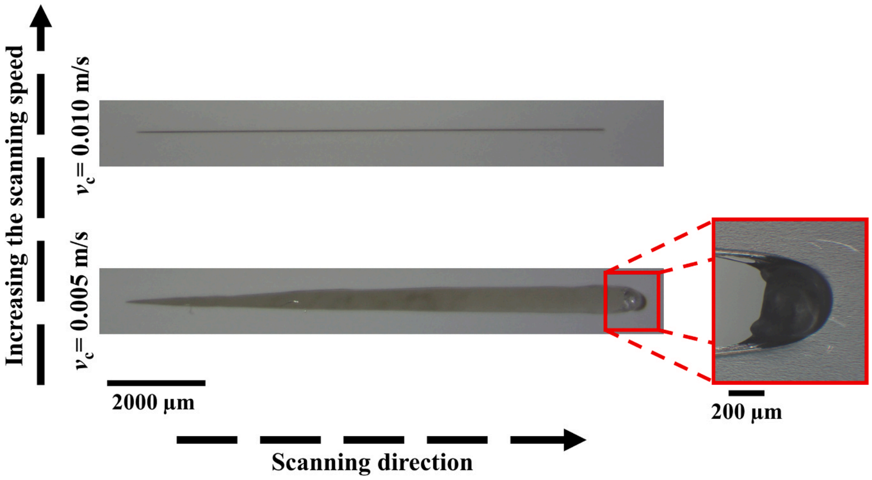

Fig. 5. Optical microscopy images of samples cut with the ns-pulsed laser system along the transverse direction (TD), using an average power of 36 W and a repetition rate of 50.0 kHz. Two distinct scanning speeds were applied to highlight the effect of scanning speed on cutting condition and thermal impact.

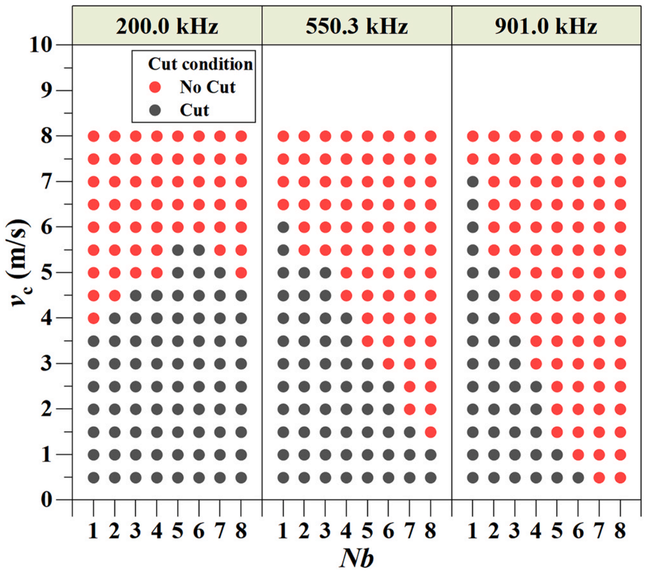

Fig. 6. The process feasibility window as a function of the pulse repetition rate, the cutting speed, and the burst mode with the ps-pulsed laser.

holes and the average value for these bulges is 6.4 µm. On the other hand, the porosities of the separators are oval in shape. These dimensions were measured for the 30 holes from the SEM images. The long dimension of the oval is along the MD and the average value is equal to 0.69 µm. Furthermore, the short dimension of the oval is along the TD and the average value is equal to 0.14 µm. As a result, the formation of small bulges is not related to the pore size. The bulges form in the direction opposite to the laser scanning. This suggests that the bulge formation is related to molten material dynamics, similar to humping in laser welding. It can likely be attributed to plastic deformation caused by plume flow. By reducing the scanning speed to 6.0 m/s, complete separation of the workpiece is visible. This scanning speed can be considered as the maximum cutting speed of the separator for the given parameter combination.

After obtaining the feasibility window for the samples with different scanning speeds, the samples with the maximum cutting speed were selected for further analysis. Fig. 8 shows the images of the samples cut at the maximum cutting speed. Fig. 9.a shows the maximum cutting speed against the different peak fluences of a single pulse. The maximum cutting speed for the ps pulse is almost three orders of magnitude higher than for the ns laser system. By increasing the peak fluence, the cutting speed increases up to a certain value. Then, by further increasing the peak fluence, the cutting speed starts to decrease. As shown in Fig. 9.b, the kerf width of the maximum cutting speed shows an increasing trend as the peak fluence is increased.

Fig. 9.c shows the fitted ablation efficiency model and the experimental data. It is possible to estimate the ablation threshold and the energy penetration depth from the regression. The apparent optical penetration depth $\widehat{\delta}$ for the separator is 6.9 µm with a 95 %

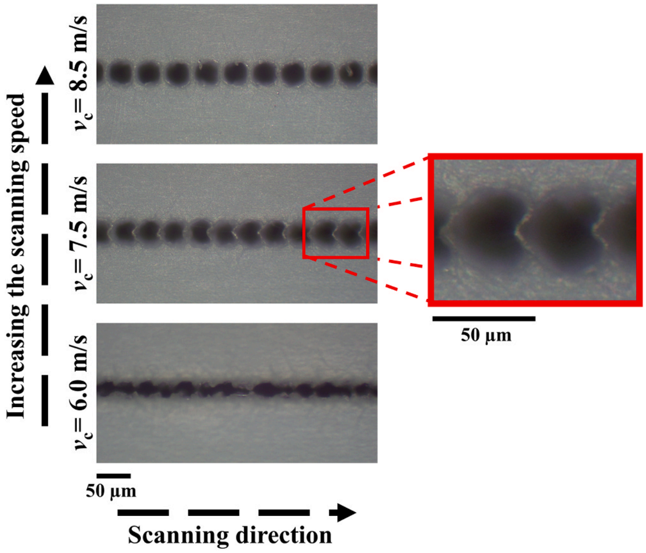

Fig. 7. Optical microscopy images of samples cut using the ps-pulsed laser system with an average power of 31 W, Nb of 6, and PRR of 200 kHz. The images illustrate the cutting conditions as the scanning speed increases.

confidence interval between [5.8, 8.2], which is very high compared to the metallic materials, remaining in the nanometric range considering optically opaque surfaces [33]. On the other hand, the ablation threshold of this material is 2.1 J/cm^2 with a 95 % confidence interval between [1.8, 2.4], which is very high compared to the metallic materials [33]. The optimum value of the peak fluence for material removal is 15.4 J/cm^2 based on the output of the ablation efficiency model. It is worth noting that the maximum cutting speed can be achieved around the optimum peak fluence.

5. Discussion

The results show that there are different cutting mechanisms at the same wavelength of 1064 nm when ns and ps pulses are used. The ns-pulsed laser produced a cut at very low scan speeds characterised by a burning-assisted melting process. The heat propagates laterally, producing an uneven kerf. The anisotropy of the material appears to play a critical role in the cutting process. The material could be cut in the transverse direction, which corresponds to the shorter side of the pores and zero thermal expansion. The material is expected to expand more freely when the laser is applied in the MD, presumably deforming the material during processing rather than causing separation. The anisotropy of the material may also produce different thermal conductivities in the two directions, giving a higher apparent thermal conductivity in the MD, resulting in no separation.

The ps-pulsed laser provides a clean cut, with holes drilled per pulse or burst packet according to the pulse repetition rate. Under these conditions, the holes drilled are wider than the porosity of the material, thus avoiding anisotropic cutting behaviour. The results did not confirm the influence of the burst mode, as the material removed per single pulse in the burst packet followed the expected trend shown by the ablation efficiency model. The fluence threshold and apparent optical penetration depth estimates allow a better

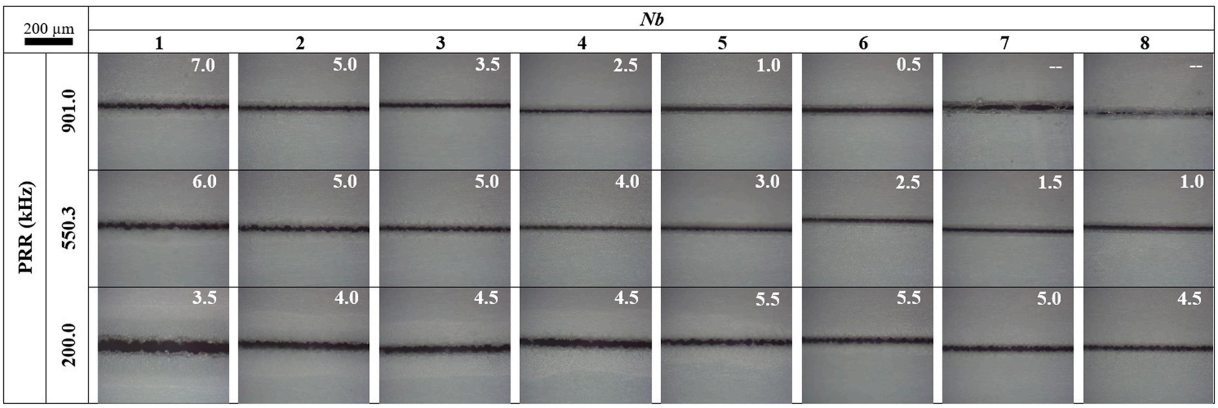

Fig. 8. Optical microscope images of the samples with the maximum cutting speed.

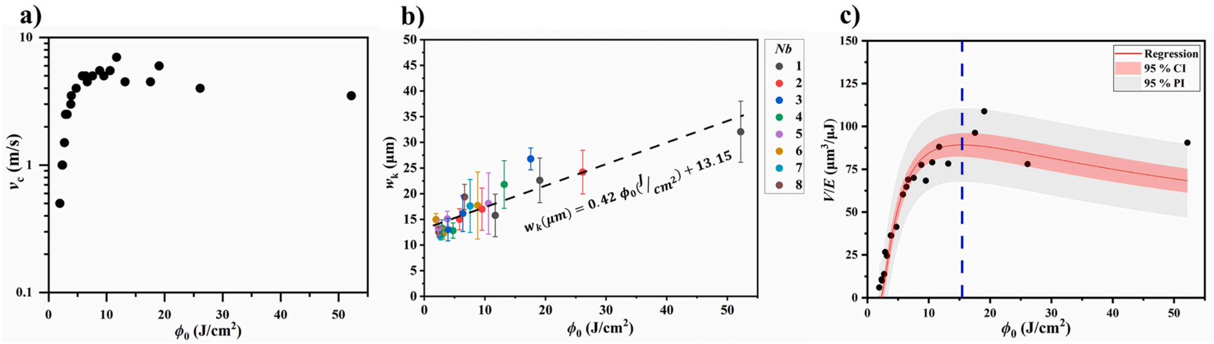

Fig. 9. a) Maximum cutting speed achieved with the ps-pulsed laser under varying peak fluence. b) Kerf width as a function of peak fluence, with the dashed line indicating the general trend. c) Fitting of the ablation efficiency model to experimental data obtained with the ps-pulsed laser system. The red shaded area represents the 95 % confidence interval for the mean, while the gray shaded area indicates the prediction interval. The dashed blue line marks the optimal fluence for maximum efficiency.

understanding of the cutting mechanism using ps pulses. The ability to cut a material that is optically transparent at 1064 nm is related to the nonlinear absorption provided by the high intensity of the ps pulses. Because transparent materials typically have large band gaps, the energy of a single photon is usually insufficient to excite electrons. However, ultrafast lasers can deliver energy within picoseconds, generating electric fields strong enough to rival the Coulomb force of atomic nuclei. This intense field enables multiphoton ionisation. At the same time, the distortion of the Coulomb potential of the valence band electrons in the direction of laser polarisation lowers the potential barrier, allowing tunnelling ionisation to occur. As more free electrons are generated by these non-linear processes, mechanisms such as impact ionisation and free carrier absorption amplify the electron population in an avalanche-like fashion. On a nanosecond timescale, this leads to a rapid rise in temperature around the laser focus, generating pressure waves that rapidly propagate through the surrounding material [37].

The ns-pulsed laser provides a peak power of in the range of 1.2–4.2 kW, while the range achieved with the ps-pulsed laser was 430–15500 kW. The experimentally observed apparent optical penetration depth can be used for further comparison with polymeric materials. The optical properties of polypropylene in the room temperature for wavelengths ranging from 0.67 up to 1.67 μm have been reported by Hakoume et al. [36]. The absorption coefficient and energy penetration depth of this material can be calculated and are equal to 0.3926 cm^{-1} and 25471 μm. On the other hand, based on the ablation efficiency model, the apparent optical absorption coefficient of the separator is calculated to be 1449 cm^{-1}, corresponding to an optical absorption depth of 6.9 μm. For opaque metals, optical penetration is limited to tens of nm at the NIR wavelength. While the apparent optical penetration is determined empirically at the μm level, the material behaves as an optically dense and non-transparent material under the influence of ps-pulsed NIR irradiation.

The optical transmission behaviour as a function of material depth is plotted shown in Fig. 10.a. The transmission behaviour with optical absorption coefficients at room temperature for fiber and CO_2 lasers at 1064 nm and 10600 nm is compared with the expected behaviour using the empirically calculated optical absorption of the ps-pulsed laser ablation. This material is completely transparent in the near infrared wavelength. However, when the ps-laser source is used at the same wavelength, the absorption changes. This means that the photon-material interaction changes when the pulse duration is shortened from ns to ps. This shows that the absorption coefficient depends not only on the wavelength but also on the pulse duration. This result shows the new possibilities that the ultrashort pulse laser system can offer in material processing of non-metallic materials. Fig. 10.b is a graphical representation of the fact that the polymer separator has a higher transmissivity when processed with the ns-pulsed laser system compared to the ps-pulsed laser system. This figure illustrates the differences in the temporal distribution of the pulses for the ns and ps pulses.

Comparing the maximum cutting speeds of the ps-pulsed and ns-pulsed laser systems, it can be concluded that the maximum cutting speed achieved by the ps laser is 700 times faster than that of the ns-pulsed laser. On the other hand, the average power of the ps laser system is 43 % lower than the average power of the ns laser system. Fig. 11 shows this point from a different perspective. This figure shows that the ps-pulsed laser system can cut the separator even with 0 % pulse overlap. This graph shows that cutting with the lowest pulse overlap occurs at the process parameter with the moderate peak fluence. On the other hand, the pulse overlap of the ns-laser system should be higher than 99.5 % to cut the sample. The cutting speed of the ns laser separator depends on the cutting direction. This is another drawback of laser-separator interaction in the melting-based process. On the other hand, the width of the cut gradually increases as the cut propagates through the sample. All these reasons show that the ps-pulsed laser system is an excellent option for scaling up laser cutting of separators in the manufacture of LIBs.

The laser system is more expensive compared to mechanical cutting in terms of the capital cost of the system. However, this cost could be justified by the flexibility to process LIB electrodes to high quality and the flexibility to process different cell sizes immediately. For mass production, this cost can be justified by the long life of the laser system and no tool wear in the long term.

Regarding the potential limitation of the ps-pulsed laser system for industrial scale up, the beam transformation from the laser source to the scanner head is one of the potential drawbacks of the ps-pulsed laser system compared to the ns-pulsed laser system. In general, the beam transformation for the ns-pulsed laser system is realised through the fiber. However, the ps-pulsed laser should be deflected by the free optics (mirrors) due to the high peak intensity for the ultrashort pulsed laser systems. This type of laser could be

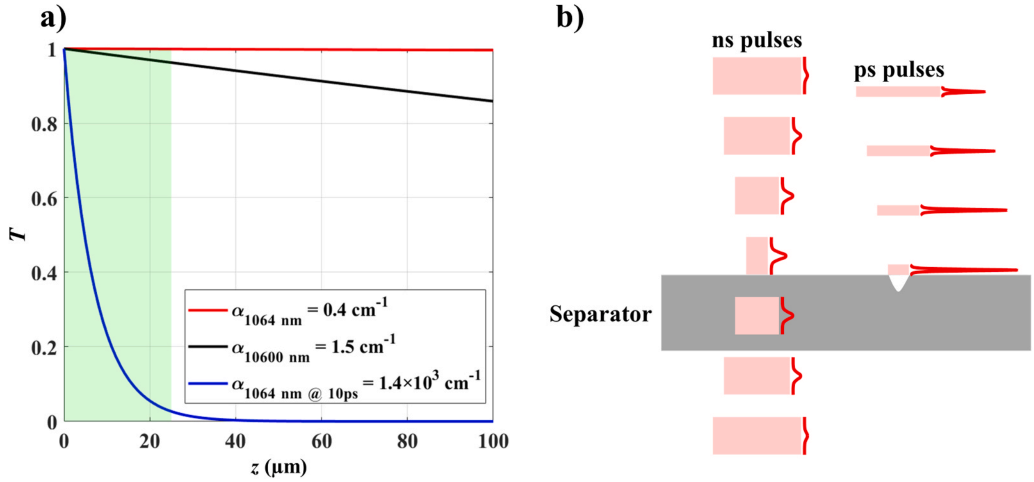

Fig. 10. a) Optical transmission as a function of length at 1064 nm and 10600 nm compared to the expected behaviour under ps pulses at 1064 nm. b) Schematic representation of LIB separator cutting with the ns and ps laser systems with near infrared wavelength, highlighting the differences in interaction mechanisms.

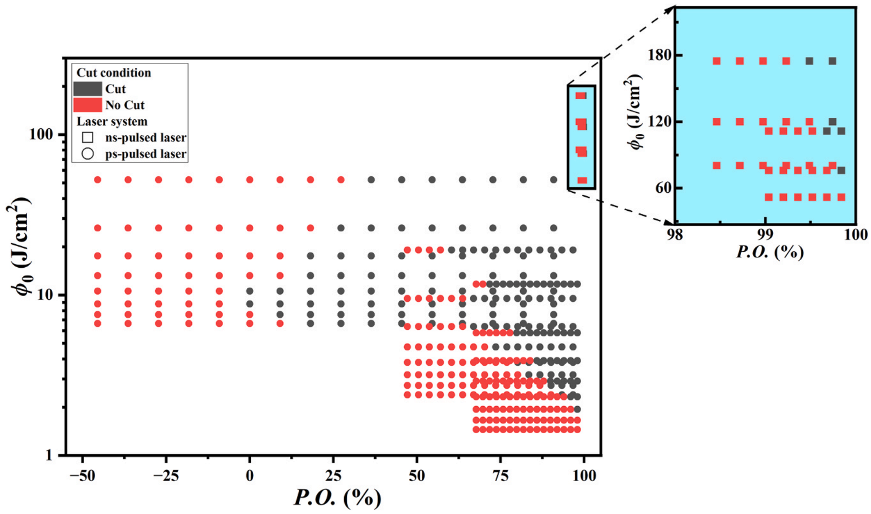

Fig. 11. Peak fluence against the pulse overlap of laser cutting with the ps- and ns-pulsed laser system.

problematic on an industrial scale due to dust contamination of the mirrors. Fiber transported ultrashort pulsed lasers are currently being developed, which are expected to resolve these issues. On the other hand, the LIB electrodes and the separator are transported in the industrial scale roll-to-roll process. The cutting of these elements in the manufacture of LIBs should be carried out while the roll is moving. For this reason, small fluctuations or vibrations of the rolls will result in movement in the vertical direction. This could cause the thin foils or separators to move out of the focal point and the intensity of the beam to be lost. This could lead to a transition from the cut state to the uncut state. To solve this problem, the mechanical structure of the industrial laser cutting station should be designed to strictly control the vibration or fluctuation of the rollers. The movement of the rollers in the vertical direction should be less than the Rayleigh length at the focal point.

For laser cutting to be comparable to mechanical cutting, the cutting speed should be more than 1 m/s. On the other hand, the wall plug efficiency of this laser source is 3.4 %. The highest cutting speed for the ps-pulsed laser system is 7 m/s. Using the cutting speed of 1 m/s in Eqs. 10 and 7 for the highest cutting speed with the ps-pulsed laser system, the average power required for cutting comparable to mechanical cutting is 4.5 W. Based on the wall-plug efficiency, the input power required is approximately 131.5 W. Despite the low wall-plug efficiency of the laser system, the laser cutting operating with a scanner head may require less energy consumption considering the fact that the mechanical cutting may require moving dies the power consumption may be relatively higher [38].

In terms of future research, the authors suggest further investigation into the application of ps- and fs-pulsed laser systems with different compositions and thicknesses. On the other hand, the wavelength of the laser source with different temporal distributions such as IR, green and UV could be interesting for further investigation for laser cutting of LIB separators.

6. Conclusions

Laser cutting of LIB separators was investigated using ps and ns laser systems. Laser cutting of the separator was performed by varying the average power, pulse repetition rate and cutting direction for the ns-pulsed laser system and by varying the number of pulses in the burst, pulse repetition rate and cutting direction for the ps-pulsed laser system.

The maximum cutting speed achievable with an ns-pulsed laser system is 0.01 m/s. Laser cutting of the separator with this system is based on melting combined with localised burning of the material. In particular, the cutting process is influenced by the cutting direction. In contrast, the ps-pulsed laser system achieves a maximum cutting speed of 7 m/s that is independent of the cutting direction, making it 700 times faster than the ns-pulsed system. At very high scanning speeds with the ps-pulsed laser system, a series of circular holes are formed during cutting. However, as the scanning speed decreases, these holes merge to form a complete separation of the workpiece.

The absorption coefficient and energy penetration depth of the material for the near infrared wavelength are 0.3926 cm^{-1} and 25471 μm, respectively. In contrast, the optical absorption coefficient derived from the ablation efficiency model is 1449 cm^{-1}. The empirical model suggests that the apparent optical penetration depth is in the micrometre range. Under ps-pulsed laser with near infrared irradiation, the material exhibits the properties of being optically dense and non-transparent. The results show that the picosecond pulsed laser at NIR wavelength can be an effective solution in the cutting of lithium-ion battery separators at high productivity and high quality. While the CO_2 lasers appear to be the more economical solution, in the upcoming years the cost of ownership the ultrafast laser systems is also expected to reduce. The ultrafast laser sources can be used effectively on other material such as metals providing an increased flexibility compared to the CO_2 systems.

CRediT authorship contribution statement

Pourya Heidari Orojloo: Investigation, Visualization, Validation, Formal analysis, Data curation, Conceptualization, Writing – review & editing, Writing – original draft, Methodology. **Ali Gökhan Demir:** Supervision, Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

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